



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Computer Networks/0705210503

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CONTROLLED ACCESS METHODS

- The stations consult one another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.

There are three popular controlled-access methods:

1. Reservation
2. Polling
3. Token passing



Reservation

- A station needs to make a reservation before sending data.

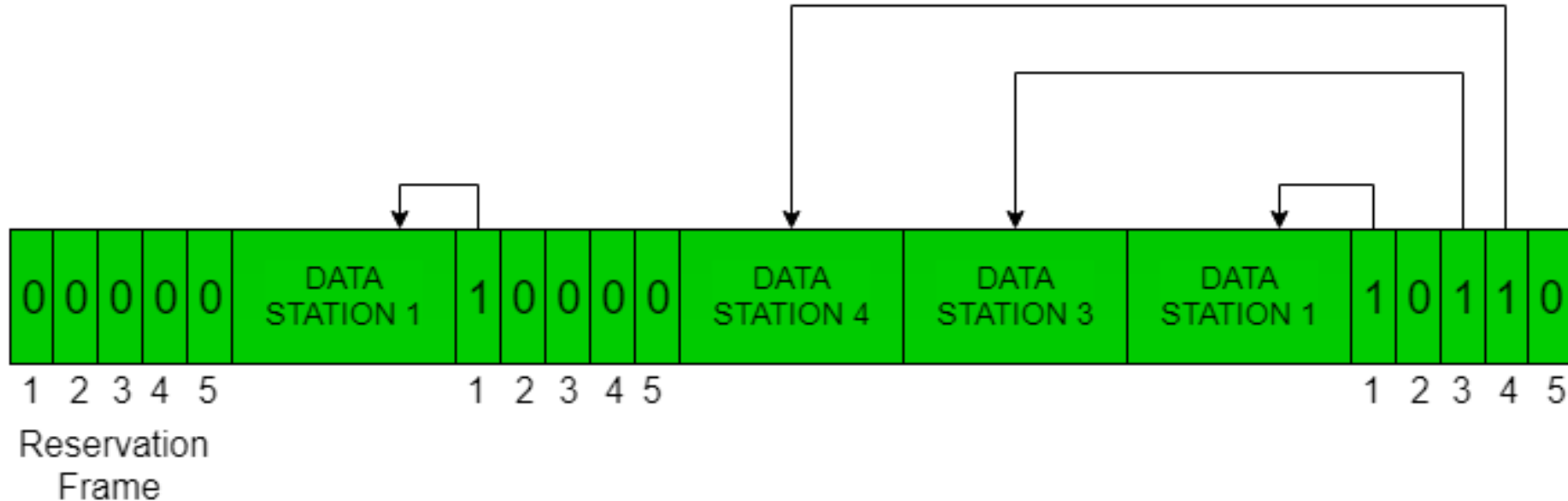
The timeline has two kinds of periods:

1. Reservation interval of fixed time length
2. Data transmission period of variable frames.

- If there are M stations, the reservation interval is divided into M slots, and each station has one slot.
- Suppose that station 1 has a frame to send, it transmits 1 bit during slot 1. No other station is allowed to transmit during this slot.
- In general, i^{th} station may announce that it has a frame to send by inserting a 1 bit into i^{th} slot.
- After all N slots have been checked, each station knows which stations wish to transmit.

Reservation

The following figure shows a situation with five stations and a five-slot reservation frame. In the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.



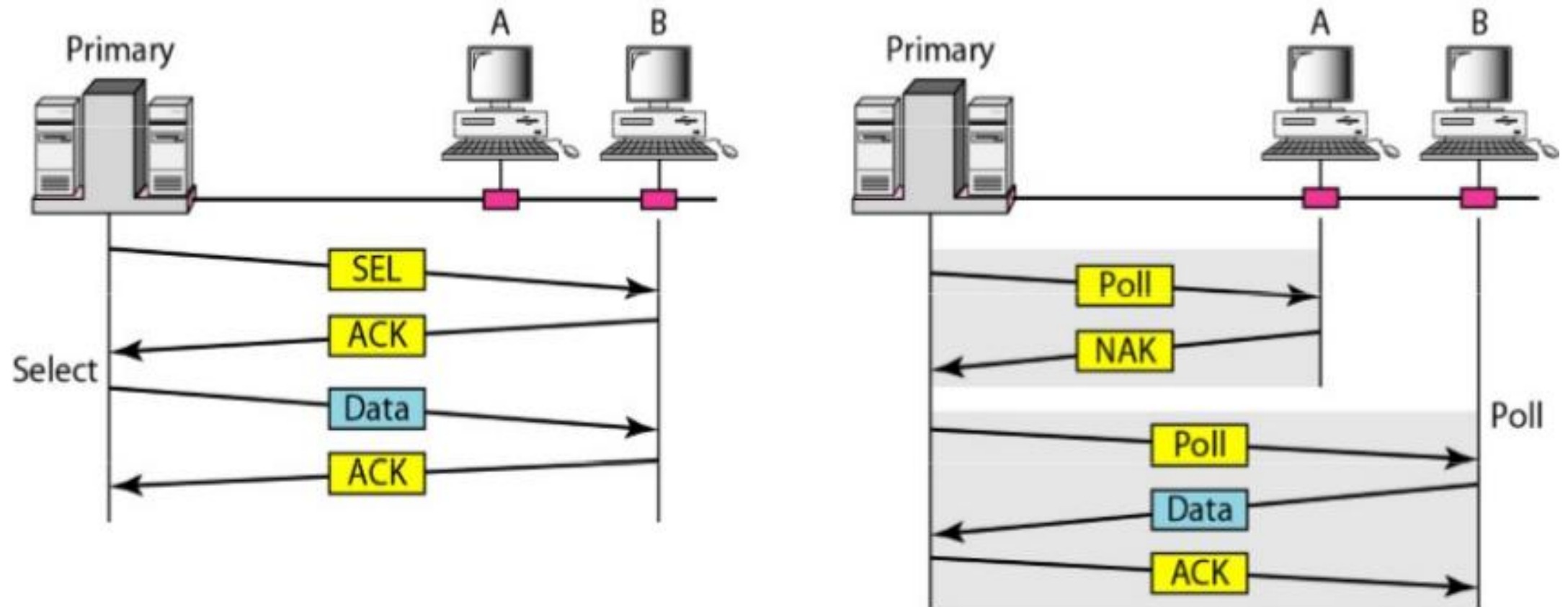


Polling

- Polling works with topologies in which one device is designated as a *primary station* and the other devices are *secondary stations*.
- All data exchanges must be made through the primary device, even when the ultimate destination is a secondary device.
- The primary device controls the link; the secondary devices follow its instructions.
- It is up to the primary device to determine which device is allowed to use the channel at a given time.
- If the primary wants to receive data, it asks the secondaries if they have anything to send; this is called **poll function**.
- If the primary wants to send data, it tells the secondary to get ready to receive; this is called **select function**.



Polling



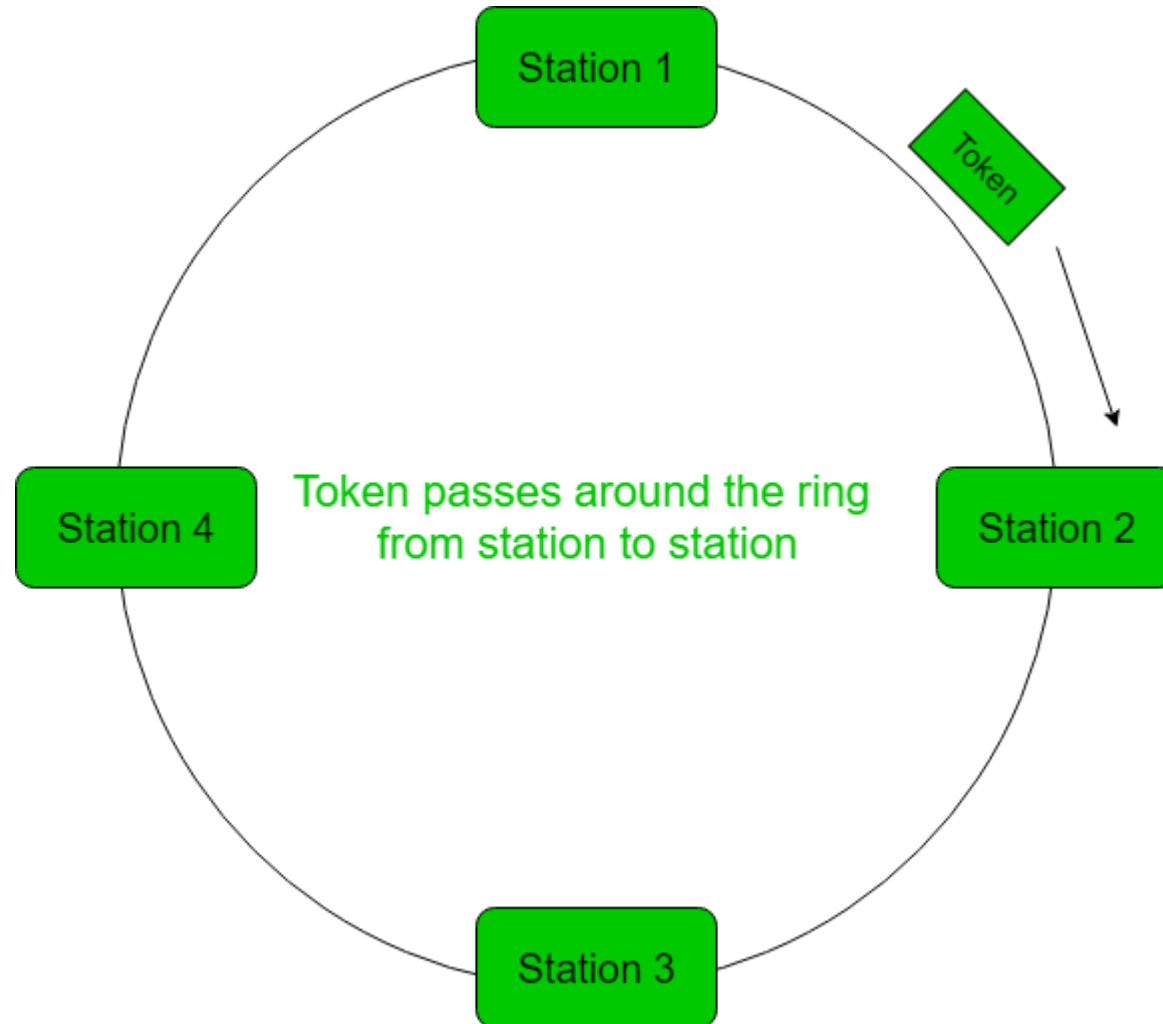


Token Passing

- The stations are connected logically to each other in form of ring and access to stations is governed by tokens.
- A **token** is a special bit pattern or a small message, which circulate from one station to the next in some predefined order.
- In Token ring, token is passed from one station to another adjacent station in the **ring** whereas incase of Token bus, each station uses the bus to send the token to the next station in some predefined order.
- In both cases, token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station. If it has no queued frame, it passes the token simply.
- After sending a frame, each station must wait for all N stations (including itself) to send the token to their neighbours and the other N - 1 stations to send a frame, if they have one.
- There exists problems like duplication of token or token is lost or insertion of new station, removal of a station, which need be tackled for correct and reliable operation of this scheme.



Token Passing





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DOCSIS (Data Over Cable Service Interface Specification)

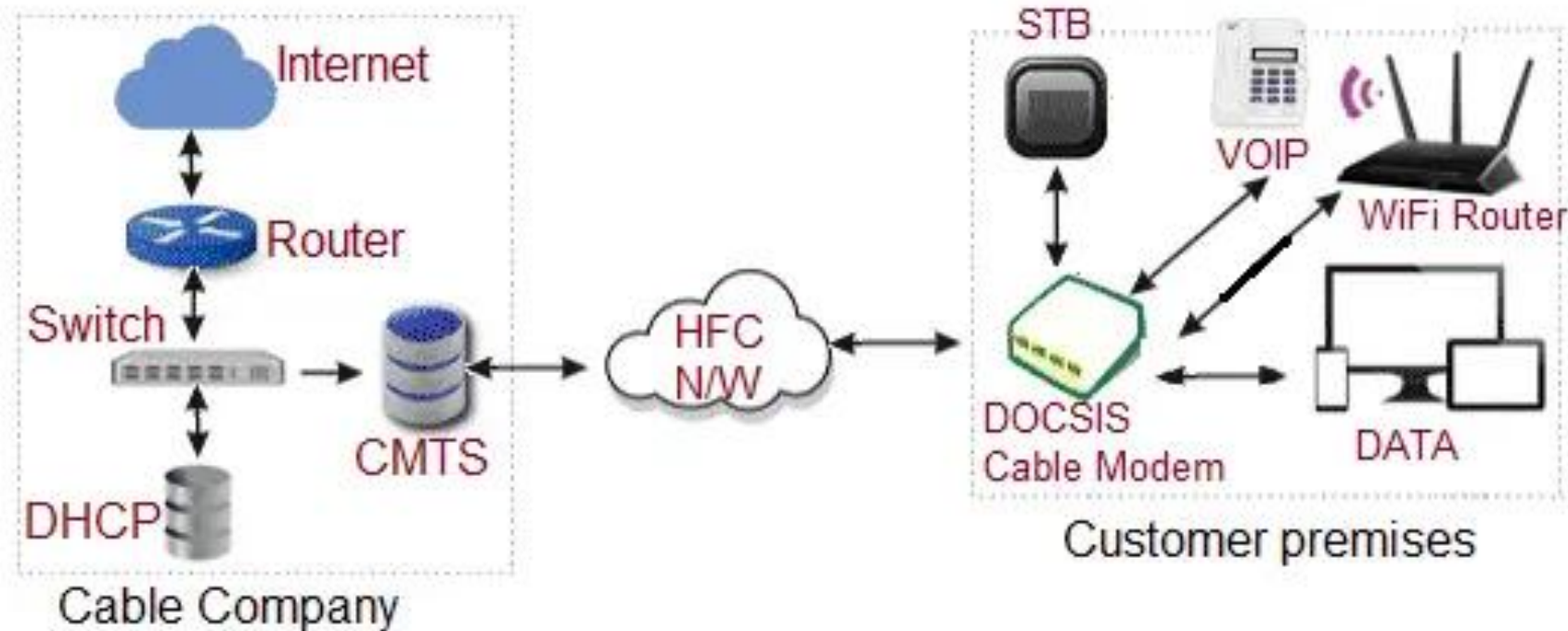
- Data Over Cable Service Interface Specification (DOCSIS) defines the interface standards for cable modems and supporting equipment involved in high speed data transfer and distribution over cable television system networks.
- Contributing companies include Cisco, Broadcom, Intel, Netgear, ARRIS, and Texas Instruments, among others.
- It allows the transmission and reception of high-bandwidth data over an existing cable TV network or HFC (Hybrid Fiber Coaxial) network.
- It permits additional high-speed data transfer over an existing cable TV system and is widely used by television operators to offer Internet access through an already existing hybrid fiber coaxial infrastructure.
- Other devices that recognize and support DOCSIS include HDTV's and Web-enabled set-top boxes for televisions.

DOCSIS (Data Over Cable Service Interface Specification)

Standard version	Features	Downstream data rate	Upstream data rate
DOCSIS 1.0	Initial cable broadband technology	40 Mbps	10 Mbps
DOCSIS 1.1	Added Voice over IP service	40 Mbps	10 Mbps
DOCSIS 2.0	Higher upstream speed	40 Mbps	30 Mbps
DOCSIS 3.0	Greatly enhances capacity	1 Gbps	100-200 Mbps
DOCSIS 3.1	Capacity and efficiency progression	10 Gbps	1-2 Gbps
DOCSIS 4.0	Symmetrical streaming and increased upload speeds	10 Gbps	6 Gbps



DOCSIS Architecture



DOCSIS System Architecture



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DOCSIS Architecture

There are three major elements of a DOCSIS system:

1. Cable modem (CM)
2. HFC network or cable TV network
3. CMTS (Cable Modem Termination System)

The DOCSIS network offers a bi-directional flow of IP data between the CM and CMTS over the existing cable network. The cable network can be either all coaxial cable or a Hybrid-Fiber/Coax (HFC) cable network.



Cable Modem (CM)

The cable modem is the main interface between the HFC network and the home network, bridging packets flowing between them. It allows connections with different customer premise equipment (CPEs) intending to access the internet.

CMTS (Cable Modem Termination System)

- It is the main element located at the cable operator's side.
- The DOCSIS standard defines messages and data types exchanged between the CMTS and the CM.
- The CMTS connects the cable operator's back office and core network to the HFC network.
- The main function of the CMTS is to forward packets between the HFC and the back office.
- It also forwards packets between upstream and downstream channels on the HFC network.
- Here, “**downstream**” refers to the data flow from the cable company to the customer premises equipment, whereas “**upstream**” refers to the data flow from the CM (i.e., cable modem) to the cable company (i.e., CMTS side).



Link Virtualization

- Link Virtualization is a network virtualization technique that allows multiple logical (virtual) links to share the same physical communication link.
- It abstracts the physical link into multiple independent virtual links, each appearing as a separate communication channel to the upper layers.
- A **physical link** → can be divided into several **virtual links**.

Each virtual link:

1. Has its own bandwidth allocation.
2. Can follow different QoS (Quality of Service) requirements.
3. Appears independent from others, even though they share the same physical medium.



Techniques of Link Virtualization

1. Multiplexing-based Virtualization

Divides the physical link based on **time** or **frequency**.

Examples:

1. **TDM (Time Division Multiplexing)** → Each virtual link gets a time slot.
2. **FDM (Frequency Division Multiplexing)** → Each virtual link gets a frequency band.

2. Packet-based Virtualization

Achieved through **packet switching**.

Uses **virtual circuits** or **labels** (like MPLS – Multiprotocol Label Switching).

3. Wavelength-based Virtualization

In optical networks → a fiber optic cable is divided into multiple virtual links using different **wavelengths**.



Techniques of Link Virtualization

4. Logical Tunneling

Encapsulation of one protocol into another (e.g., VPNs). Creates virtual links over existing IP networks.

Applications

- VPNs (Virtual Private Networks) → Secure tunnels over shared internet.
- MPLS networks → Virtual paths for guaranteed QoS.
- Optical networks → Wavelength virtualization for high-capacity transmission.
- Cloud computing & Data centers → Dynamic allocation of bandwidth to tenants.